STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 8: Mechanical Properties of Solids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Define stress. [1]
2. SI unit of Young's modulus? [1]
3. Longitudinal strain? [1]
4. Hooke's law? [1]
5. Modulus for volume change? [1]
6. Is strain dimensionless? [1]
7. Meaning of large Young's modulus? [1]
8. Longitudinal stress? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. 100 N on 2x10^-4 m^2: stress? [2]
10. 2 m wire extends 1 mm: strain? [2]
11. Stress=2x10^7 Pa, strain=10^-4: Y? [2]
12. 10 cm cube compresses by .1 cm: strain magnitude? [2]
13. L=1 m,A=2x10^-6 m^2,Y=2x10^11,F=40 N: extension? [2]
14. Why use steel reinforcement in concrete? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. Wire L=2m,A=10^-6,Y=2x10^11,F=100N: extension? [3]
16. Stress 4x10^7 Pa at strain 2x10^-4. Estimate modulus. [3]
17. B=1.5x10^9 Pa, volume strain=-2x10^-4. Pressure increase? [3]
18. Why does doubling wire length double its extension if F,A,Y fixed? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Distinguish proportional and elastic limits. [4]
20. Derive extension and calculate for F=50 N,L=1.5m,A=10^-6,Y=2x10^11. [4]

End of paper - check working, units, signs and significant figures.